Cryptography sheet # 4

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In this sheet, we reflect about the relation of what we have been doing in the lecture to groups and try to understand what makes a public-key cryptosystem reasonable. We also get enough courage to start analyzing the structure of the power functions for arbitrary rings $\mathbb{Z}_n$. Last but not least, we learn a neat way to differentiate between rings and fields.

1 Power functions in finite groups

You have seen that Fermat's little theorem is a useful result that allows us to invert power functions over the field $\mathbb{Z}_p$ for a prime p. It is a common intention in quantitative sciences to make results more general by expanding their scope (that is, simplifying the assumptions) or by making the assertions stronger. So, let us try to make a step towards generalizing Fermat's little theorem just a little more. We don't have to strive for the most general version one knows, though. Let's just take the setting of an Abelian group $(G, \cdot)$ with the group operation written as a multiplication. Assume, our group has a finite size $k = |G| \in \mathbb{N}$ (or, as would one usually say in the group-theory jargon, G is a finite group of order k). Show that $a^k = 1$ holds for every $a \in G$. Here, 1 is the (standard) notation for the neutral element of $(G, \cdot)$. Remark: there is an even more general version of this result, but sometimes it might be overwhelming to digest the most general version, so that it makes sense to let the degree of generality grow slowly.

2 BeanCrypt

Mr. Bean got a brilliant idea for a public-key cryptosystem. According to his newly devised system BeanCrypt, the public key consists a prime number p with lots of digits, cosets $a \in \mathbb{Z}_p \setminus \{0\}$, $b \in \mathbb{Z}_p$ and a natural number e satisfying $\gcd(e, p-1) = 1$. The public key should be used by a sender for encryption of a plain text $x \in \mathbb{Z}_p$ as the cipher text $\mathrm{f}(x) = ax^e + b$. The private key (u, v, d, p) determines the function $\mathrm{g}(x) = (ux + v)^d$ inverse to $\mathrm{f}(x)$. The owner of the private key decrypts $y = \mathrm{f}(x)$ by applying g to y. Mr. Bean submits BeanCrypt for consideration to the patent office. Imagine that you are a patent examiner and you are somewhat sick (to say the least) because you are confronted with a huge number of crappy patents flooding in, which makes it challenging for you to review applications thoroughly. Try to explain to Mr. Bean without cursing why BeanCrypt is not quite good. Elaborate on the computational efficiency in your explanation.

3 Caesar's new idea

Caesar has close friends in the patent office and they told him about Mr. Bean's patent application that uses the function $\mathrm{f}(x) = ax^e + b$. While the idea of using this kind of functions over $\mathbb{Z}_p$, where

p is prime, in the way described by Mr. Bean looks crazy, one can still try using the function f in symmetric cryptography, when both parties share the common key (a, e, b) and keep it private. Caesar wants to use f to reshuffle the symbols of his favorite alphabet, which is the Latin one (what a surprise!). There is a problem, though. He'd need to know what exponents $e \in \mathbb{N}$ he could use. Obviously, he cannot try all the infinitely many exponents. Fermat told him that if the number of letters in the alphabet were prime, then Caesar would be working in the field $\mathbb{Z}_n$ and could restrict e to the values from 1 to $n-1$, as $x^n = x$ for every $x \in \mathbb{Z}_n$, which implies that the functions in the infinite list $x^1, x^2, x^3, \ldots$ repeat themselves periodically. But Caesar's n is not a prime number, it is $n = 26$. If periodicity of power functions over $\mathbb{Z}_n$ is true for a general $n \geq 2$, then one could impose an upper bound on e. Help Caesar figure out if the periodicity is true for any $n \geq 2$. To make the statement of the problem rigorous, we introduce additional terminology. We call an infinite sequence of objects $s_1, s_2, s_3, \ldots$ eventually periodic if there exist $m \in \mathbb{N}$ and $t \in \mathbb{N}$ such that $s_i = s_{i+t}$ when $i \geq m$. You need to show, for every $n \geq 2$, that the sequence of the power functions is eventually periodic.

4 A generalization of Wilson's theorem

A unitary commutative ring $(R, +, \cdot)$ is called field if every nonzero element of R is invertible. How can we decide whether a given finite unitary commutative ring $(R, +, \cdot)$ is a field? A natural way to do this would be to calculate all products $a \cdot b$ with $a, b \in R$ and to test if for every a at least one of the products $a \cdot b$ evaluates to 1_R. But this seems to be rather inelegant. Indeed, there is a better criterion for answering this question.

Let $(R, +, \cdot)$ be a finite commutative unitary ring with neutral elements 0_R and 1_R. Show that $(R, +, \cdot)$ is a field if and only if:

$$\prod_{x \in R \setminus \{0_R\}} x = -1_R.$$

Hint 1: If R is a field, is there any $x \in R \setminus \{-1_R, 1_R\}$ such that $x^2 = 1_R$?

Hint 2: Suppose a and b are elements of R and a is a zero divisor. What can we conclude about the product $a \cdot b$?

Bonus exercise: We know that $\mathbb{Z}_n$ is a field if and only if n is a prime number. Using the above, can we say anything about the value of $(n-1)! \mod n$ depending on whether or not n is a prime number?